

Adoption and use of digital technologies in everyday life and organisational contexts

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Content of the lecture

- Internet use and non-use in Sweden
- Technology adoption models
- Social shaping of technology
- Domestication theory
- Uses and gratifications theory
- Application areas in health and welfare technology



"Technologies need to be used to have any impact. People do not automatically start using technology or robots; individuals' behaviours and actions towards any technology will be affected by each person's priorities, assumptions, expectations and understandings of the technology in question, such as perceptions about whether a technology can help them attain their goals or not" (Frennert et all., 2021; Martínez-Torres et al., 2015)

Swedes and the Internet (annual report 2024)

- A unique survey of the internet use in Sweden (Swedish Internet Foundation)
- Internet usage has neither increased nor decreased in 2024: Almost 95 % of Swedes use the internet
- Older women are among the lowest proportion of internet users
- One in ten Swedes of working age need digital assistance
- The need for digital assistance quadruples in retirement age
- The "pre-digital generation" needs assistance with most things online
- https://svenskarnaochinternet.se/rapporter/svenskarna-och-internet-2024/

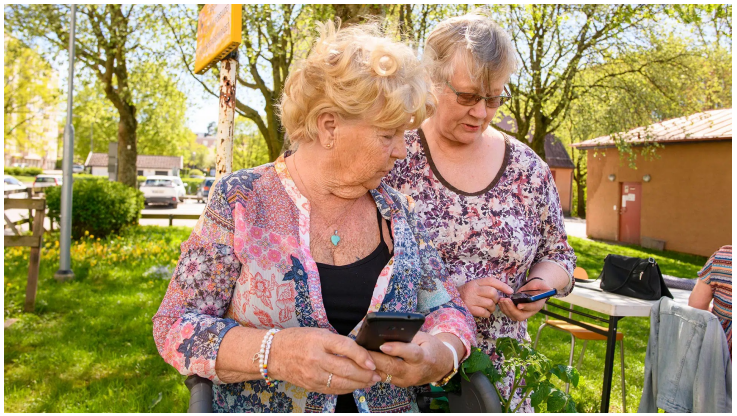
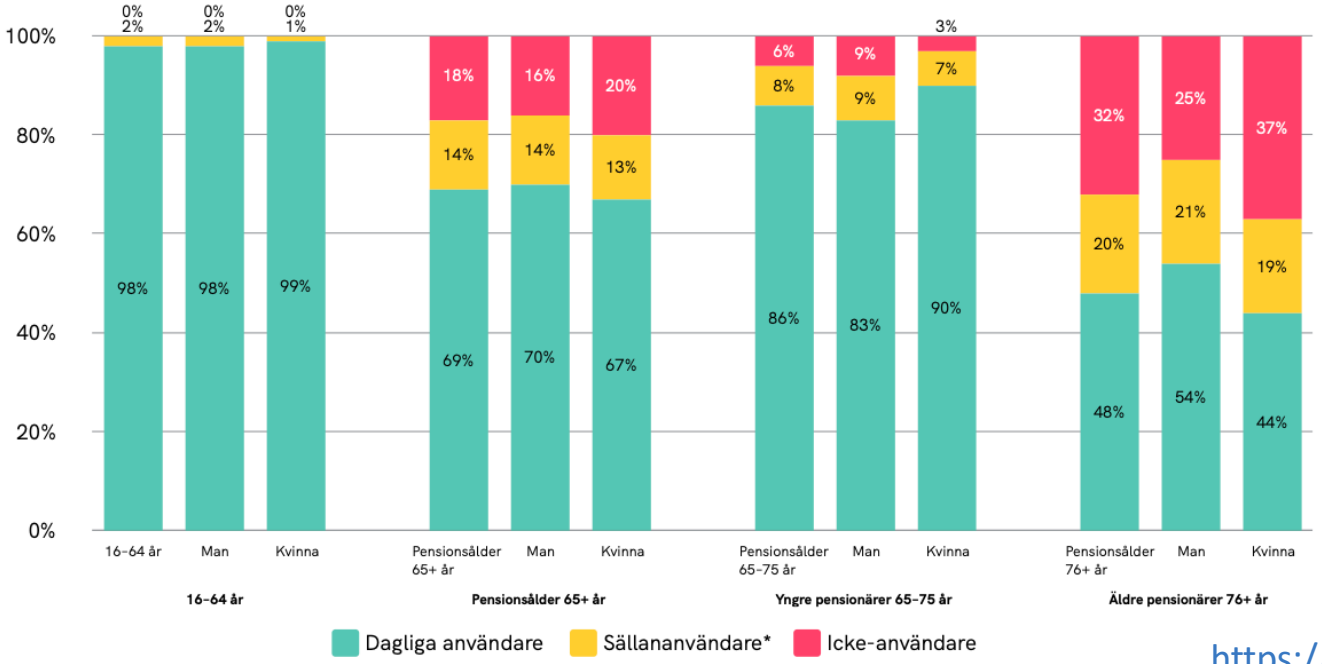


Image: Internetstiftelsen


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Use and non-use of internet in Sweden (2024)

Fråga: Hur ofta använder du internet?



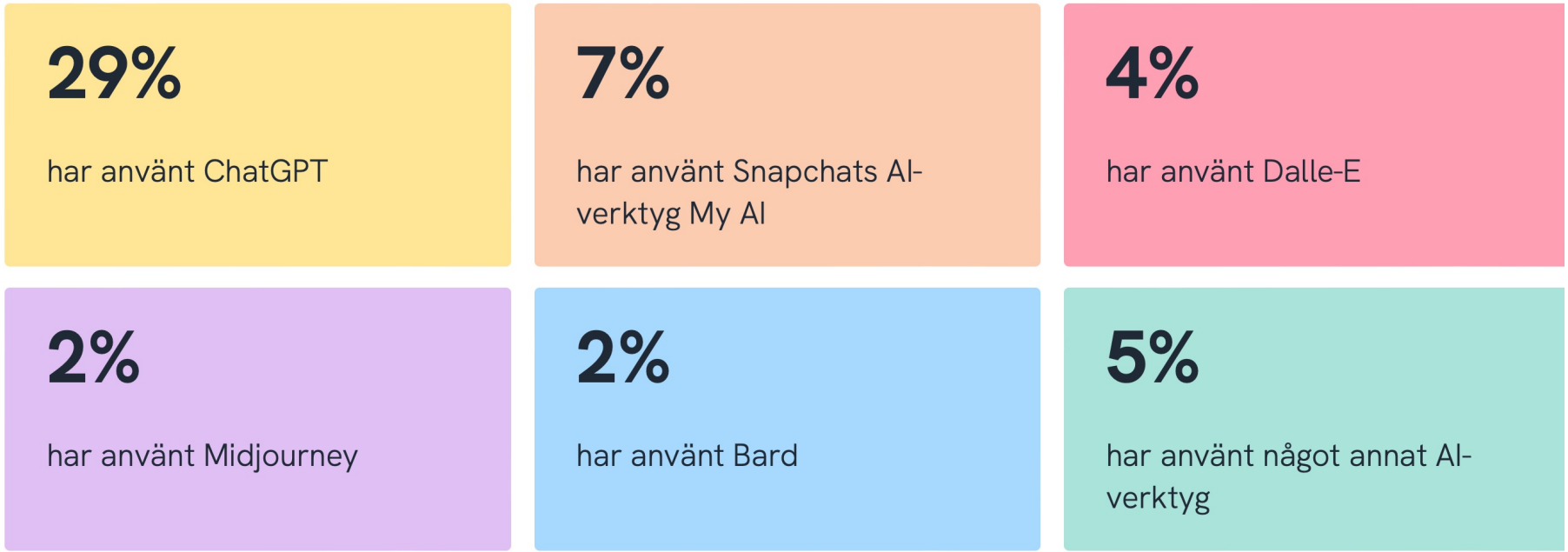
Those who are not using the internet are adults aged 65 and above

Diagram 1.2, Bas: Samtliga 16+ år, År 2024 (Internetpenetrationsstudie)

*Med sällananvändare avses de internetanvändare som inte använder internet dagligen.

https://svenskarnaochinternet.se/rapporter/svenskarna-och-internet-2024/

Swedes and AI (annual report 2024)



Internetstiftelsen (2024). Svenskaran och AI.
https://svenskarnaochinternet.se/utvalt/svenskarna-och-ai-2024/

Swedes and AI (annual report 2025)

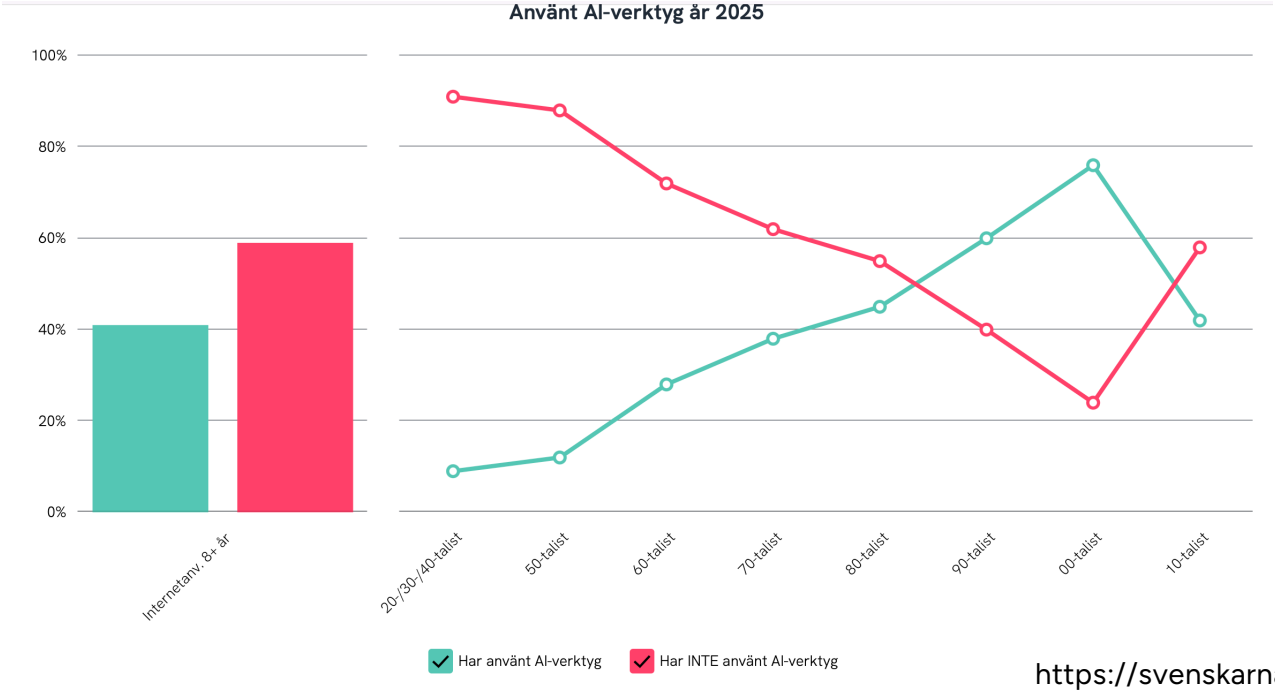


Diagram 2.1a, Bas: Internetanvändare 8+ år, År 2025 (Studie 1) För exakt frågeformulering, se frågeformulär.

Young adults (born in the 00s) are using AI tools most frequently

https://svenskarnaochinternet.se/rapporter/svenskarna-och-internet-2025/ai-artificiell-intelligens/

Positive turn in attitudes towards care robots (Turja et al. 2022)

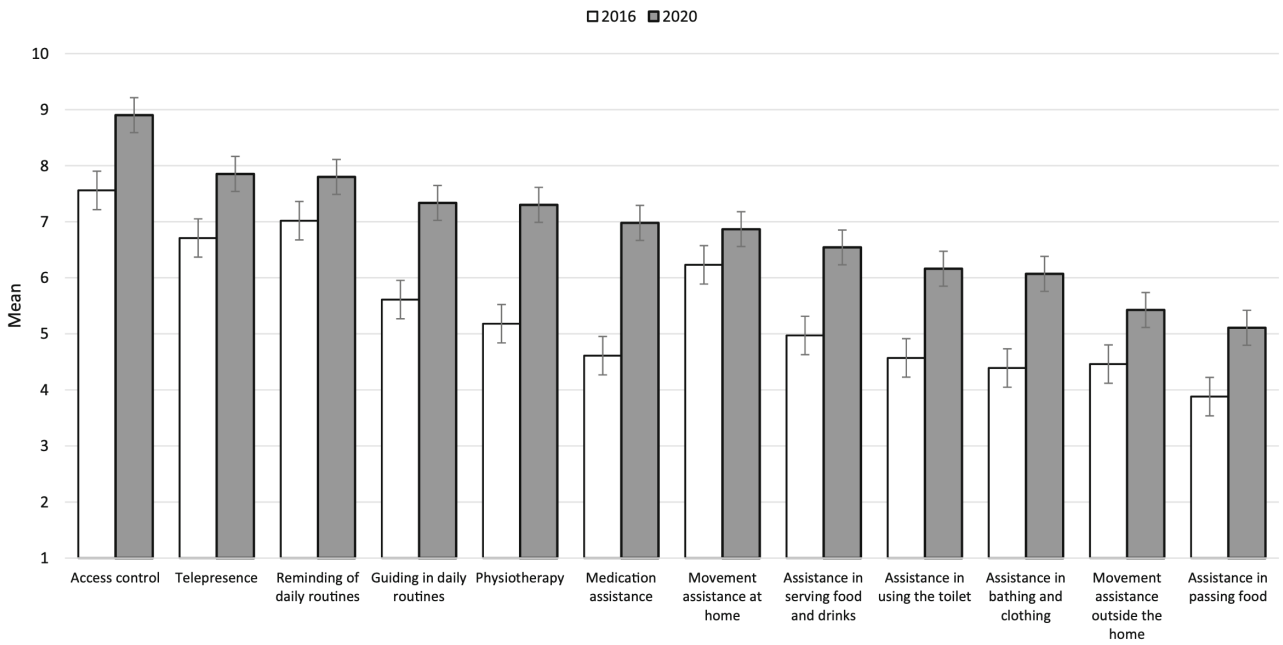


Fig. 1 Perceived usefulness of telecare robots, means per scenario, per year

Turja, T., Taipale, S., Niemelä, M., & Oinas, T. (2022). Positive Turn in Elder-Care Workers' Views Toward Telecare Robots. *International Journal of Social Robotics*, *14*(4), 931-944. https://doi.org/10.1007/s12369-021-00841-2



Technology adoption models (TAM, UTAUT)

- TAM – Technology acceptance model (Davis, 1989)
- ***Perceived usefulness*** - The degree to which a person believes that using a particular system would enhance their job performance. Perception of whether or not someone perceives that technology to be useful for what they want to do.
- ***Perceived ease-of-use*** – The degree to which a person believes that using a particular system would be free from effort. If the technology is easy to use, then the barriers conquered.
- ***External variables*** such as social influence

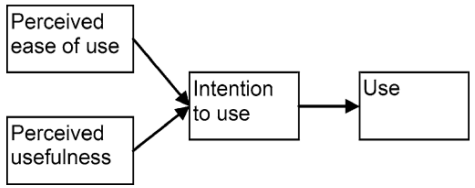


Fig. 2 Basic TAM assumptions

- UTAUT - Unified theory of acceptance and use of technology (Venkatesh et al., 2003)
- Determinants of usage intention and behaviour:
- 1) performance expectancy, 2) effort expectancy, 3) social influence and 4) facilitating conditions.
- Gender, age, experience, and voluntariness of use are posited to moderate the impact of the four key constructs on usage intention and behavior.

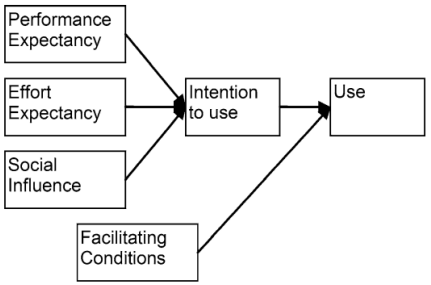


Fig. 3 UTAUT model: direct influences

The Almere model for robot acceptance among older adults

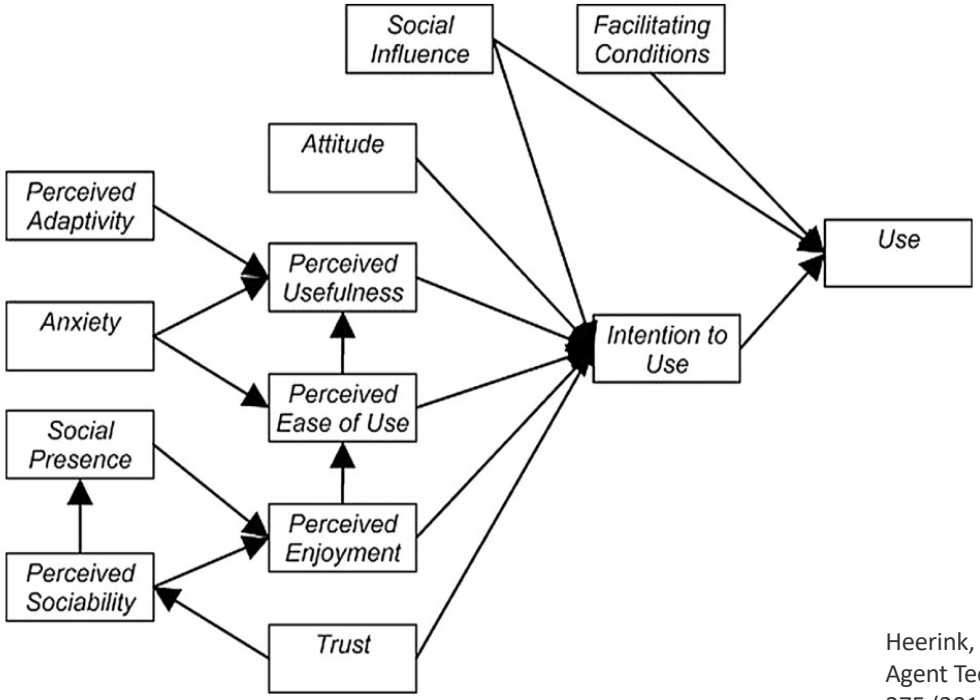


Fig. 1. The Almere model of robot acceptance among older adults [29].



Heerink, M., Kröse, B., Evers, V. *et al.* Assessing Acceptance of Assistive Social Agent Technology by Older Adults: the Almere Model. *Int J of Soc Robotics* **2,** 361–375 (2010). https://doi.org/10.1007/s12369-010-0068-5

Almere-model in usability evaluation

User experience	Male	Female	Under 76	76 and over
Sociability				
Robot was a pleasant conversational partner	3.08	3.33	3.40	3.00
Conversation with the robot was similar to conversation with family members	2.00	1.87	1.73*	2.15*
Robot can give me similar social support than friends and family members	2.15	1.73	1.80*	2.08*
Social presence				
Robot felt like a real person	2.00	1.87	1.87	2.00
Usefulness				
Robot can help me to reduce the experience of loneliness	3.15	2.93	3.07	3.00
Ease-of-use				
It was easy to understand the robot	4.23	4.40	4.27	4.38
It was easy to understand how to talk with the robot	4.31	4.13	4.07	4.38
It was easy to begin the conversation with the robot without addional help	4.15*	3.47*	3.80*	3.77*
Comprehension				
Robot could understand and hear me	3.92	3.33	3.73	3.46
Intention to use				
I would like to use robot in my own home	2.38	2.73	2.67	2.46
Adaptiveness				
Robot can help me when I perceive it as necessary	2.92	3.07	2.73	3.31
Robot will give me unnecessasry advice	2.23	2.73	2.47	2.54
Turn-taking				
Robot could understand when I wanted to begin the conversation	3.46	3.60	3.47	3.62
Robot did not interrupt me	4.15	3.80	3.87	4.08
Security				
It would be safe to use the robot in my home	3.38	3.47	3.60	3.23
Trust				
I can trust the robot	2.85	3.30	2.73	3.38

Kuoppamäki, S., Irfan, B., McCarren, L.: Reciprocity in dialogue with conversational companion robots with older adults (in preparation)

Social Shaping of Technology (SST)

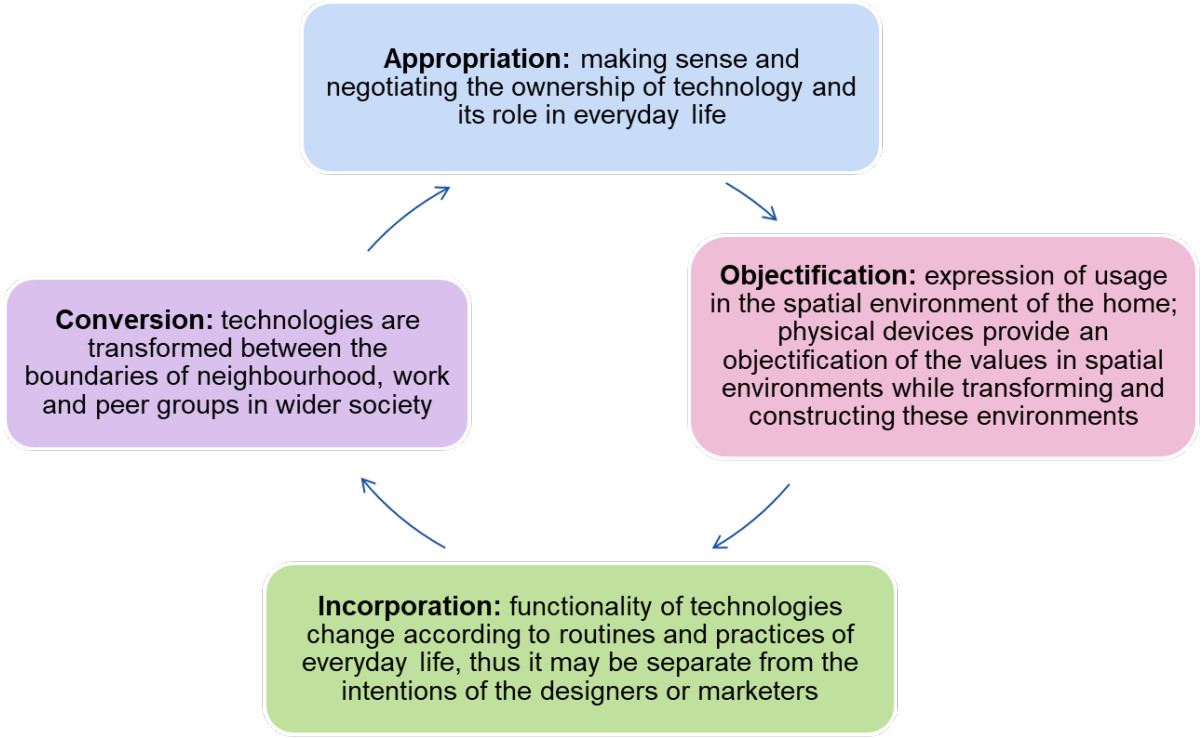
- MacKenzie, Donald and Wajcman, Judy, eds. (1999). Introductory essay: the social shaping of technology. http://eprints.lse.ac.uk/28638/1/Introductory%20essay%20%28LSERO%29.pdf
- Criticism of *technological determinism*: assumption that technologies develop linearially based on scientific advance or following a logic of their own, outside human or societal influences
- Technology shapes technology
- Economic shaping of technology
- Economic shaping is social shaping
- Mutual shaping between technology and society: Not only technology affects society, but society affects technology through economics, politics, and culture
- Connection to Social construction of technology and Actor-Network Theory (Latour)





Domestication of technology (Silverstone & Haddon, 1996)

- People construct their experience with technologies through social meaning-making processes
- Technologies influence the social and economic order of the household where they connect family members
- Within the moral economy of the household, i.e. the economic unit of its members, households are engaging with the products and their meanings



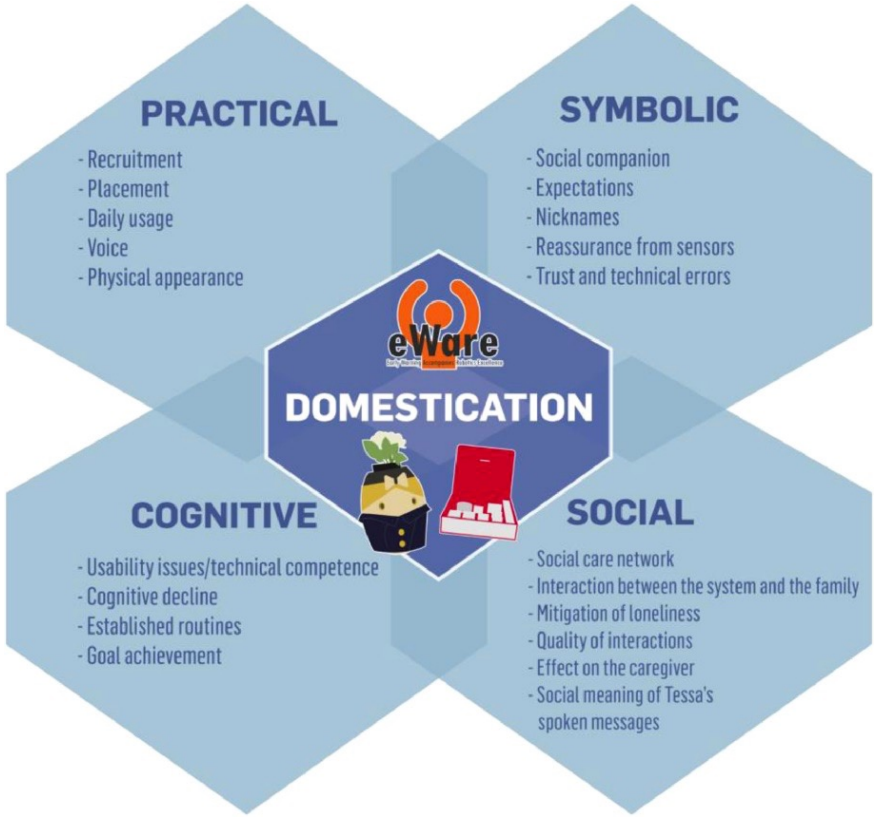
Example: Domestication of drones



- Example: Drones as a part of everyday life
- Appropriation: Making sense of drones, their functionality, purpose
- Objectification: Interpreting drones as part of everyday life, household structure, social environment
- Incorporation: Shaping and constructing the meaning of drones so that they are fitted for each individual purpose
- Conversion: Technologies are becoming part of everyday life, thus shaping individuals' practices, habits, routines

Example: Interactions between older adults, caregivers, and robots in the home (Søraa et al., 2021)

- Technology is used – domesticated – in different practical, symbolic and cognitive ways.
- Domestication of technology theory should be expanded to include a social dimension.
- Robot technology can have multiple effects on care networks and relationships.



Søraa, R.A., Nyvoll, P., Tøndel, G., Fosch-Villaronga, E., Serrano, J.A. (2021). The social dimension of domesticating technology: Interactions between older adults, caregivers, and robots in the home. Technological Forecasting and Social Change. https://doi.org/10.1016/j.techfore.2021.120678

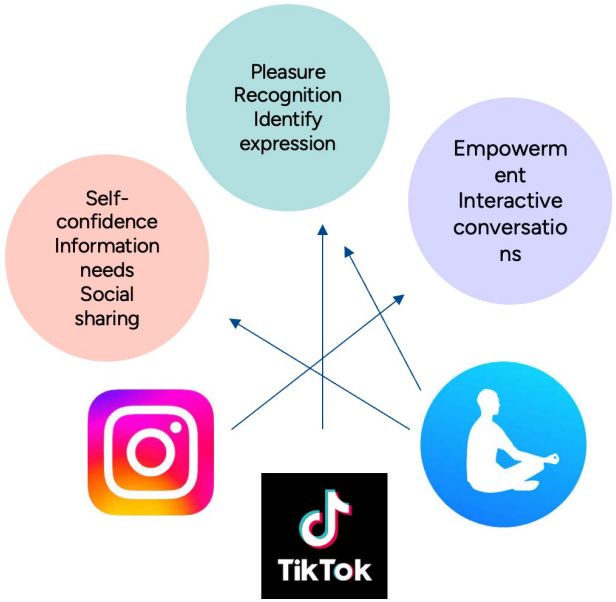
Different levels of domestication Søraa et al., 2021)

- ***Practical domestication***: Physical, observable interactions that users have with technology. How technology is bought, placed, or used in the physical sense?
- ***Symbolic domestication***: Unobserved consequences of adopting the technology, i.e., what it means or symbolises for the user to have the technology in their life?
- ***Cognitive domestication:*** How the users learn from and through technology and the practices relating to the use of the technology. How is technology enacted, and what change in practices does this entail?



Uses and gratifications theory (UGT)

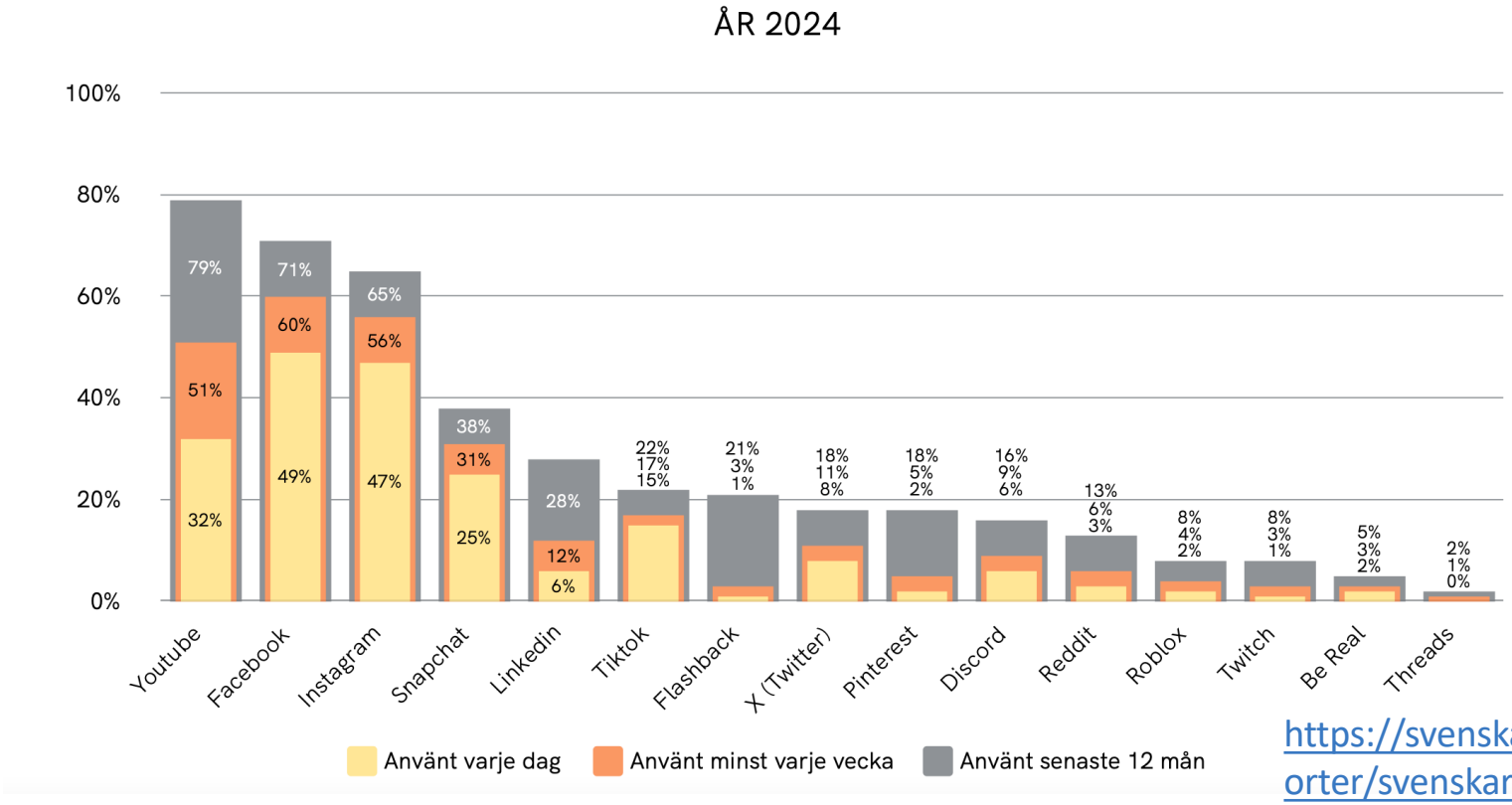
- Approach to understanding why and how people actively seek out specific technology or media to satisfy specific needs
- Audience-centered approach to understand the use of information and communication technologies
- What do people do with media? Why do they use food apps? Why young adults use social media and for what purposes?
- Socio-psychological understanding of the communication tradition: Users actively choose what media technology they use based on their specific needs
- Users are not only passive consumers of media
- Digital technology use is influenced by physical, social and emotional needs



Uses and gratifications theory (UGT)

Social media use in Sweden (2024)

Fråga: Vilka sociala nätverksplatser/sociala medier har du använt de senaste 12 månaderna?



https://svenskarnaochinternet.se/rapporter/svenskarna-och-internet-2024/

Perceived benefits and drawbacks in social media use

Fråga: Vilka av följande påståenden instämmer du i?

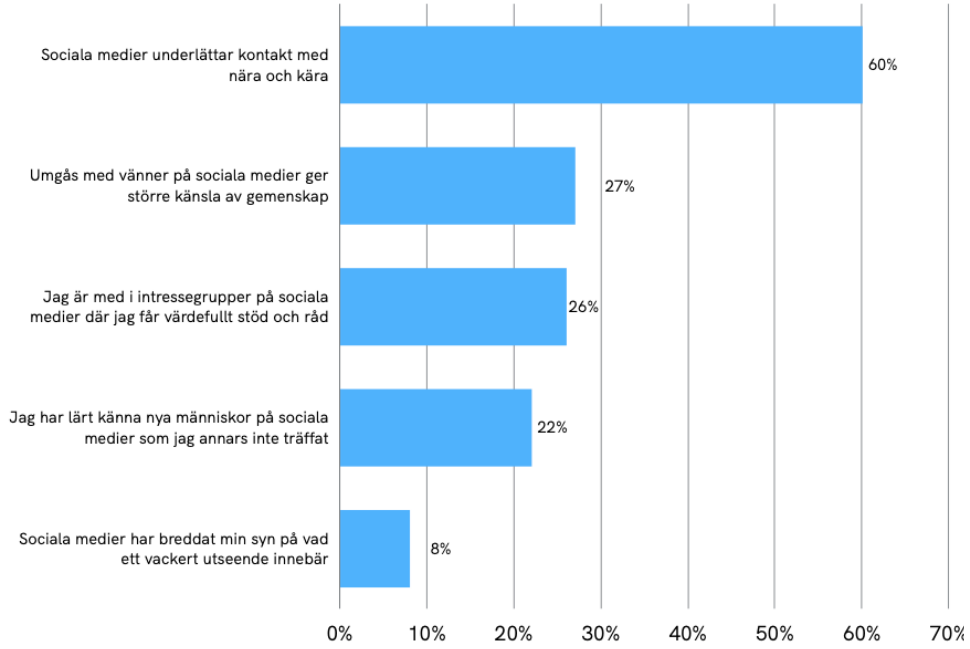


Diagram 3.9, Bas: Internetanvändare 12+ år, År 2024 (Studie 1)

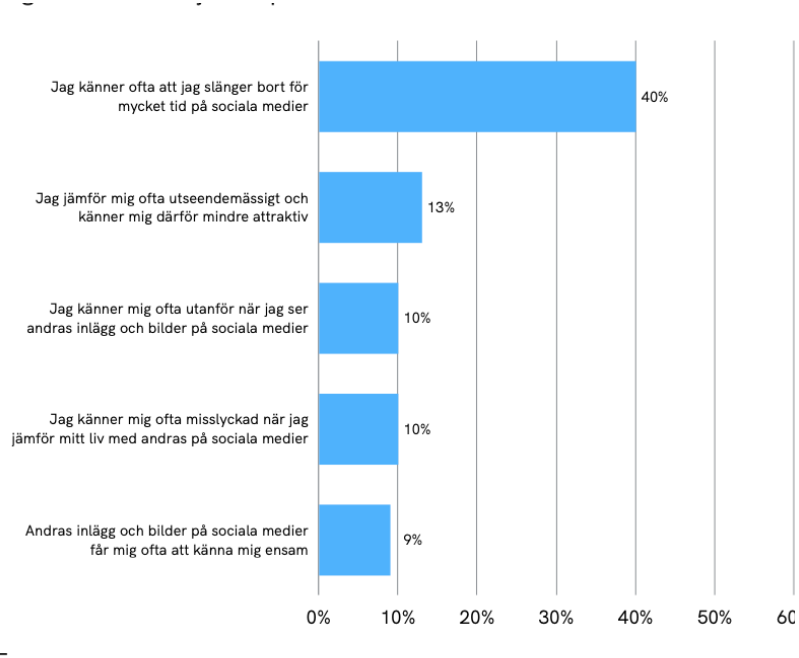


Diagram 3.15, Bas: Internetanvändare 12+ år, År 2024 (Studie 1)

https://svenskarnaochinternet.se/rapporter/svenskarna-och-internet-2024/

Break 15 min!



Group discussion

1) Choose one **example of medical/health technology** that is relatively new in organisational contexts, such as in hospitals, municipal care facilities, nursing homes, rehabilitation centers (e.g., technology that is not yet widely used)
2) Describe different stages of domestication of this particular technology
3) What are the possible challenges of domestication/adoption of this technology in this context?
4) Document your discussion: https://padlet.com/sannaku/ygyr2f164u48p544







Examples of technology adoption

Appropriation: Making sense of digital mindfulness

- Breathing exercises facilitating improved bodily awareness
- Decrease in muscle tension
- Improvement in breathing technique (e.g. holding one's breath)
- Relaxation
- Positive thinking and self-reflection

You get irritated and frustrated, and you get angry at yourself, you are so frustrated by making a mistake. (--) Why did I say this? And then I realise that maybe you can handle it in a different way (--) That maybe I can look at my well-being and my inner self in a different way (--). I often get stressed by something I have done wrong. (P10, 73, female)

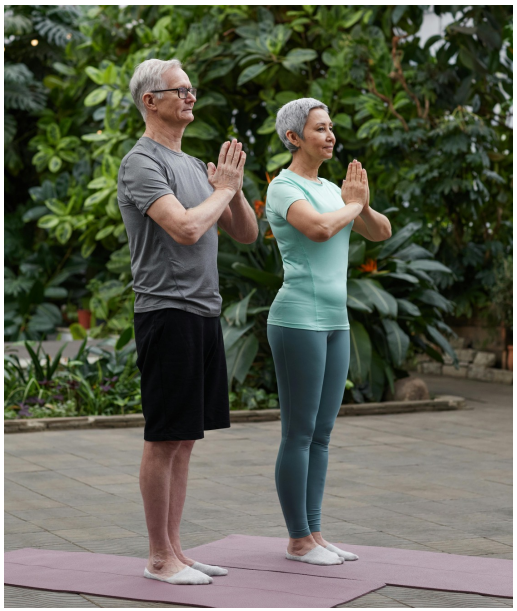


Image: Pexels

Objectification: Intepreting mindfulness in daily life

- Improvement in sleep through breathing exercises
- Alleviation of pain
- Supporting mental health
    - Learning how to manage mental health conditions (e.g. symptoms of ADHD)
    - Learning how to accept age-related changes

I've had PMR (polymyalgia rheumatica) for a year now, and it has been so hard (--), there is a lot of pain sometimes. When you sit and just feel that pain, or lie down and just feel that, then it's very difficult to get your mind off it. But with the mindfulness app, I disconnect from that. So, then the pain disappears. (P12, 78, female)



Image: Pexels

Incorporation – Shaping the meaning of mindfulness

- Digital mindfulness improved flexibility and accessibility of training
- Temporality of the app use (morning vs. evening)
- Familiarity of locations and timing improved adherence
- Challenges to adhere to new mindfulness routines

I liked that it took five minutes. I almost always did it in the morning. And then I did it while lying in bed. So, I was in the same place. (--) I tried it (once) when I was waiting for a bus. But I found that just disturbing. So that was the good thing about the app. It took five minutes, a routine. (P3, 73, female)



Image: Pexels

Conversion: Digital mindfulness shaping well-being

- Mindfulness training can have negative effects (e.g. emotional sensitivity, traumatic re-experiencing)
- Physical discomfort due to lack of experience in digital mindfulness
- Emotional distress due to associating mindfulness with negative life events

Breathing is emotionally charged for me because my brother passed away from ALS (Amyotrophic Lateral Sclerosis), which makes it hard to breathe. So it made me think about that. I've had quite a bit of pain at times in my life—not really now—but it also reminded me of that. (P1, 68, female)



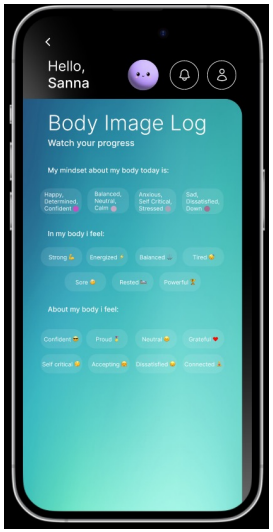
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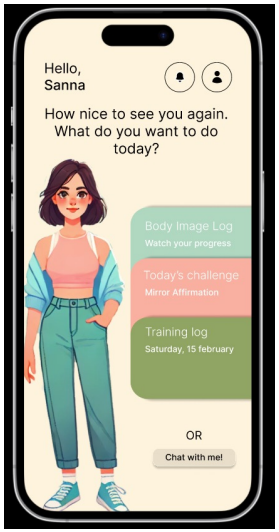
Key take away message

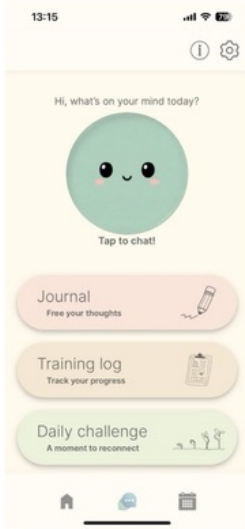
Technology adoption is always connected to the person's social environment, previous technology use, and personal meanings associated with the technology

Understanding users to inform new designs

Understanding user's insights, perspectives, and experiences can result to a more user-centered technology designs







Preparations for tomorrow (Seminar 26/3)

- Purpose is to present and discuss the current stage of the user study, give and receive feedback to other's work and raise any questions in different stages of the user study
- You need to be prepared to present the *preliminary idea* for your user study (technology, product, context, user group)
- 15 min per group (10 min presentation, 5 min discussion)
- You can work individually, pairs or in small groups
- Please use power points when presenting

